# Lunar Descent Plant and Flight Software: Mathematics

## 1 Conventions

SI units throughout: meters, seconds, kilograms, radians. Frame: origin at the Moon's center, $z$ along the Moon's spin axis, axes fixed relative to the stars (Moon-centered inertial). The Moon does not move. Source: `workloads/moon/common/`.

## 2 Vectors

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ and scalar $s$:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{1}$$

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x) \tag{2}$$

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} \tag{3}$$

The cross product follows the right-hand rule: $\hat{x} \times \hat{y} = \hat{z}$. Code: `vec3.hpp`. Tests: `test_vec3.cpp`.

## 3 Constants

| Symbol | Value | Unit | Source |
|---|---|---|---|
| $\mu_M$ | $4.902800066 \times 10^{12}$ | $\mathrm{m}^3/\mathrm{s}^2$ | JPL DE430 |
| $R_M$ | 1 737 400 | m | mean radius |
| $\omega_M$ | $2.6616995 \times 10^{-6}$ | rad/s | sidereal spin, one turn per 27.321661 days |
| $g_0$ | 9.80665 | $\mathrm{m}/\mathrm{s}^2$ | standard gravity |

The spin vector is $\boldsymbol{\omega}_M = (0, 0, \omega_M)$. The ground under a point $\mathbf{r}$ moves at $\boldsymbol{\omega}_M \times \mathbf{r}$, which is $\omega_M R_M = 4.624$ m/s at the equator. Code: `constants.hpp`.

## 4 Gravity

With $r = \|\mathbf{r}\|$:

$$\mathbf{g} = -\mu_M \frac{\mathbf{r}}{r^3} \tag{4}$$

At the surface, $\mu_M / R_M^2 = 1.6242$ m/s$^2$. Earth's tidal acceleration near the Moon is on the order of $10^{-5}$ m/s$^2$ (estimated) and is not modeled. Code: `dynamics.cpp`. Tests: `test_dynamics.cpp`.

## 5 Engine and actuator limits

The flight software commands a throttle $\tau$ and a direction $\mathbf{d}$. The plant applies the limits of a real engine:

$$\tau' = \begin{cases} 1 & \tau > 1 \\ \tau & 0 < \tau \leq 1 \\ 0 & \text{otherwise (including NaN)} \end{cases} \tag{5}$$

The engine is off ($\tau' = 0$) when $\|\mathbf{d}\|$ is zero, infinite, or NaN, or when the mass has reached the dry mass $m_{\text{dry}}$. Otherwise:

$$F = \tau' F_{\max} \tag{6}$$

$$\mathbf{a}_T = \frac{F}{m} \frac{\mathbf{d}}{\|\mathbf{d}\|} \tag{7}$$

$$\dot{m} = -\frac{F}{I_{sp}\, g_0} \tag{8}$$

The direction is normalized, so a corrupted command can point the engine anywhere but cannot push harder than full thrust. These limits are physics of the plant, not a flight software defense.

## 6 Equations of motion

$$\dot{\mathbf{r}} = \mathbf{v} \tag{9}$$

$$\dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}) + \mathbf{a}_T \tag{10}$$

$$\dot{m} = -\frac{F}{I_{sp}\, g_0} \tag{11}$$

The system does not depend on time explicitly. Code: `dynamics.cpp`, function `derivative`.

## 7 Numerical integration (RK4)

Write the state as $\mathbf{y} = (\mathbf{r}, \mathbf{v}, m)$ and the equations of motion as $\dot{\mathbf{y}} = f(\mathbf{y})$. The classical fourth-order Runge–Kutta step of size $h$:

$$\mathbf{k}_1 = f(\mathbf{y}_n) \tag{12}$$

$$\mathbf{k}_2 = f(\mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1) \tag{13}$$

$$\mathbf{k}_3 = f(\mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_2) \tag{14}$$

$$\mathbf{k}_4 = f(\mathbf{y}_n + h\,\mathbf{k}_3) \tag{15}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{h}{6}\,(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \tag{16}$$

The engine command is held constant for the whole step, as flight software sets one command per tick. If $m_{n+1} < m_{\text{dry}}$, it is set to $m_{\text{dry}}$; the step still applied thrust for its full length, so the error is at most one step of thrust.

**Validation.** With no thrust, specific orbital energy $\mathcal{E} = \frac{1}{2}\|\mathbf{v}\|^2 - \mu_M/r$ and angular momentum $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ are conserved. Over one 100 km circular lunar orbit (period 7067.5 s) in 5000 steps, measured drift is $2.1 \times 10^{-15}$ in $\mathcal{E}$ and $8.9 \times 10^{-16}$ in $\|\mathbf{h}\|$, at the level of double rounding, and the spacecraft returns to within a millimeter of its start. Forward Euler on the same orbit drifts $1.5 \times 10^{-2}$ in $\mathcal{E}$ and returns 138 km off. Code: `integrator.cpp`. Tests: `test_integrator.cpp`.

# 8 Navigation (flight software)

The flight software receives the Moon-centered state $(\mathbf{r}, \mathbf{v})$ and derives the ground-relative state:

$$\mathbf{u} = \mathbf{v} - \boldsymbol{\omega}_M \times \mathbf{r} \qquad \text{velocity relative to the ground} \tag{17}$$

$$\hat{\mathbf{n}} = \mathbf{r}/\|\mathbf{r}\| \qquad \text{local up} \tag{18}$$

$$h = \|\mathbf{r}\| - R_M \qquad \text{altitude} \tag{19}$$

$$v_\uparrow = \mathbf{u} \cdot \hat{\mathbf{n}} \qquad \text{vertical speed, positive up} \tag{20}$$

$$v_\leftrightarrow = \|\mathbf{u} - v_\uparrow \hat{\mathbf{n}}\| \qquad \text{horizontal speed} \tag{21}$$

This stage runs before guidance, so a fault here reaches every TMR replica.

**Limitations.** The real lunar spin axis is tilted about 1.5° from the orbit normal (Cassini's laws); the model fixes it along $z$. Altitude is above a sphere of radius $R_M$; there is no terrain. Code: `fsw/navigation.cpp`. Tests: `test_navigation.cpp`.

# 9 Guidance (flight software)

Inputs: the navigation state and the mass $m$. Output: throttle $\tau$ and direction $\mathbf{d}$; the plant applies the actuator limits.

**Braking.** $\tau = 1$, $\mathbf{d} = -\mathbf{u}/\|\mathbf{u}\|$: full thrust against the ground-relative velocity.

**Switch.** Braking ends, permanently, when $\|\mathbf{u}\| < 50$ m/s or $h < 2000$ m, whichever comes first.

**Final descent.** Target vertical speed and wanted acceleration:

$$v^* = -\operatorname{clamp}\left(\frac{h}{10\ \text{s}},\ 1,\ 30\right)\ \text{m/s} \tag{22}$$

$$g = \frac{\mu_M}{(R_M + h)^2} \tag{23}$$

$$\mathbf{a}^* = \left(g + k_v\left(v^* - v_\uparrow\right)\right)\hat{\mathbf{n}} - k_h\left(\mathbf{u} - v_\uparrow \hat{\mathbf{n}}\right) \tag{24}$$

with $k_v = 1.0\ \text{s}^{-1}$ and $k_h = 0.5\ \text{s}^{-1}$. If $\mathbf{a}^* \cdot \hat{\mathbf{n}} \le 0$ the engine is off (a lander cannot thrust toward the ground). Otherwise

$$\tau = \frac{\|\mathbf{a}^*\|\, m}{F_{\max}}, \qquad \mathbf{d} = \mathbf{a}^* \tag{25}$$

and the plant saturates $\tau$ at 1. On the target descent rate with no drift, $\tau = g\, m/F_{\max}$, the hover throttle.

**Initial values.** The switch thresholds, the 10 s time constant, the 1 and 30 m/s limits, and both gains are starting values, to be tuned against the first full landing. Code: `fsw/guidance.cpp`. Tests: `test_guidance.cpp`.